

Year 8 Mathematics 2019

Mini-Test 3- Rates, Ratios, Decimals & Percentages

Resource: Calculators and 1 A4 page of note



Time: 35 mins

Name: Answers

/32 marks

Part A : Multiple Choice (1 mark per question)

For all questions in Part A circle the correct answer.

Question 1 (1 mark)

What is 25% of \$440?

You may use this side of the page for any working out

- a) \$110
- b) \$270
- c) \$170
- d) \$50

Question 2 (1 mark)

Which of the following is 11% of \$220?

- a) \$11.10
- b) \$12.10
- c) \$231
- d) \$24.20

Question 3 (1 mark)

If I increase \$55 by 10% I get which of the following:

- a) \$59.10
- b) \$61.50
- c) \$60.50
- d) \$55.10

Question 4 (1 mark)

If I decrease \$36 by 50% I get which of the following: Circle the correct answer.

- a) -\$14
- b) \$86
- c) \$61
- d) \$18

Question 5 (1 mark)

If my cost price is \$41 and my selling price is \$44, what is my profit?

a) \$3

b) -\$9

c) \$3

d) \$13

Either

Question 6 (1 mark)

If a bike worth \$450 is sold at a loss of 20%, its sale price is:

a) 540

b) 405

c) 360

d) 430

Question 7 (1 mark)

3 cm : 6 cm simplified is:

a) 9 : 1

b) 2 : 1

c) 1 : 2

d) 6 : 1

Question 8 (1 mark)

Divide \$60 in the ratio of 4:2

a) \$20: \$40

b) \$40: \$20

c) \$6: \$10

d) \$8: \$12

Question 9 (1 mark)

A bread recipe uses 250 g of flour to 500 ml of water. If I use 100 g of flour then the volume of water I should use is:

a) 500 ml

b) 250 ml

c) 200 ml

d) 850 ml

Question 10 (1 mark)

Maria can touch type 150 words in 3 minutes. How many words can she type in 1 minutes?

- a) 50 words
- b) 20 words
- c) 30 words
- d) 150 words

Question 11 (1 mark)

If Brad travels at 80km/h on his moped for 4hrs, how far will he go

- a) 20km
- b) 80km
- c) 240km
- d) 320km

Question 12 (1 mark)

The fraction $\frac{27}{55}$ written in exact decimal form is:

- a) $0.\overline{490}$
- b) $0.\dot{4}\dot{9}\dot{0}$
- c) $0.4\overline{90}$
- d) $0.\dot{4}9\dot{0}$

Part B: Short Answer Question

Question 1 (2 marks)

Use an appropriate notation to simplify writing these recurring decimals. (2 marks)

a) $1.33333333333333 =$

$1.\dot{3}$ ✓
or $1.\overline{3}$

b) $0.0656565 =$

$0.0\overline{65}$ or $0.0\dot{6}\dot{5}$ ✓

Question 2 (2 marks)

On a test Steve was to answer the question $7 \div 11$. On the screen of his calculator is the following;

0.63636363636363

He has written his answer as $0.\dot{6}$. Is Steve correct? Justify your answer.

$0.\overline{63}$ or $0.\dot{6}\dot{3}$ ✓

Explanation of why ✓

Question 3 (4 marks)

a) A prepaid mobile phone deal offers 29c per text message, what would be the charge for 83 text messages?

29×83 ✓
 $= \$24.07$ ✓

b) The same deal offers 43c per 30 seconds for phone calls, what would be the charge for a 3 min call?

$30 \times 6 = 3 \text{ mins}$ ✓

$43 \times 6 = \$2.58$ ✓

Question 4 (2 marks)

Use a calculation to determine which of the following are the best buys:

a) \$12.45 for 500 g of chocolate or \$15.05 for 750 g?

$12.45 \div 5 = \$2.49 / 100g$

$15.05 \div 7.5 = \$2.01 / 100g$

Any appropriate comparison units are acceptable ✓

Best Buy = \$15.05 for 750g ✓

Question 5 (4 marks)

a) Simplify the following ratios:

i) $4 : 8$

$1 : 2 \quad \checkmark$

ii) $0.2 : 1.2$

$$2 : 12 \quad \left(\frac{1}{2}\right) \text{ marks for this}$$

$$= 1 : 6 \quad \checkmark \text{ full marks for this.}$$

b) The concrete is to be made of cement, sand, and gravel in the ratio of $1 : 2 : 3$.

If the mixer can take 30 shovels before it overflows, how many shovels of sand should I use when making a batch of concrete?

$1 : 2 : 3$

$1 + 2 + 3 = 6 \text{ parts}$

$30 \div 6 = 5 \text{ shovels per part} \quad \checkmark$

$50 \quad 5 : 10 : 15$

therefore 10 shovels of sand. $\checkmark$ **Question 6 (6 marks)**

Michelle bought 10 shirts for \$15 each.

a) If she expects to sell these for \$20 each, what is the percentage profit she will make on each shirt?

$20 - 15 = \$5 \text{ profit} \quad \checkmark$

$\% \text{ profit} = \frac{5}{15} \times 100 = 33.\bar{3} \% \quad \checkmark$

b) She sells 8 of the T-shirts at \$20, but the final two T-shirts she sells for \$12. What is the percentage loss on the final two T-shirts?

$12 - 15 = \$3 \text{ loss} \quad \checkmark$

$\% \text{ loss} = \frac{3}{15} \times 100 = 20 \% \quad \checkmark$

c) What is the total percentage loss or profit Michelle made overall for selling all 10 T-shirts?

$8 \times 20 = 160$

$$2 \times 12 = 24$$

$$\hline 184 \text{ revenue}$$

$10 \times 15 = \$150 \text{ cost}$

$184 - 150 = \$34 \text{ profit} \quad \checkmark$

$$\frac{34}{150} \times 100 = 22.\bar{6} \% \text{ profit} \quad \checkmark$$

